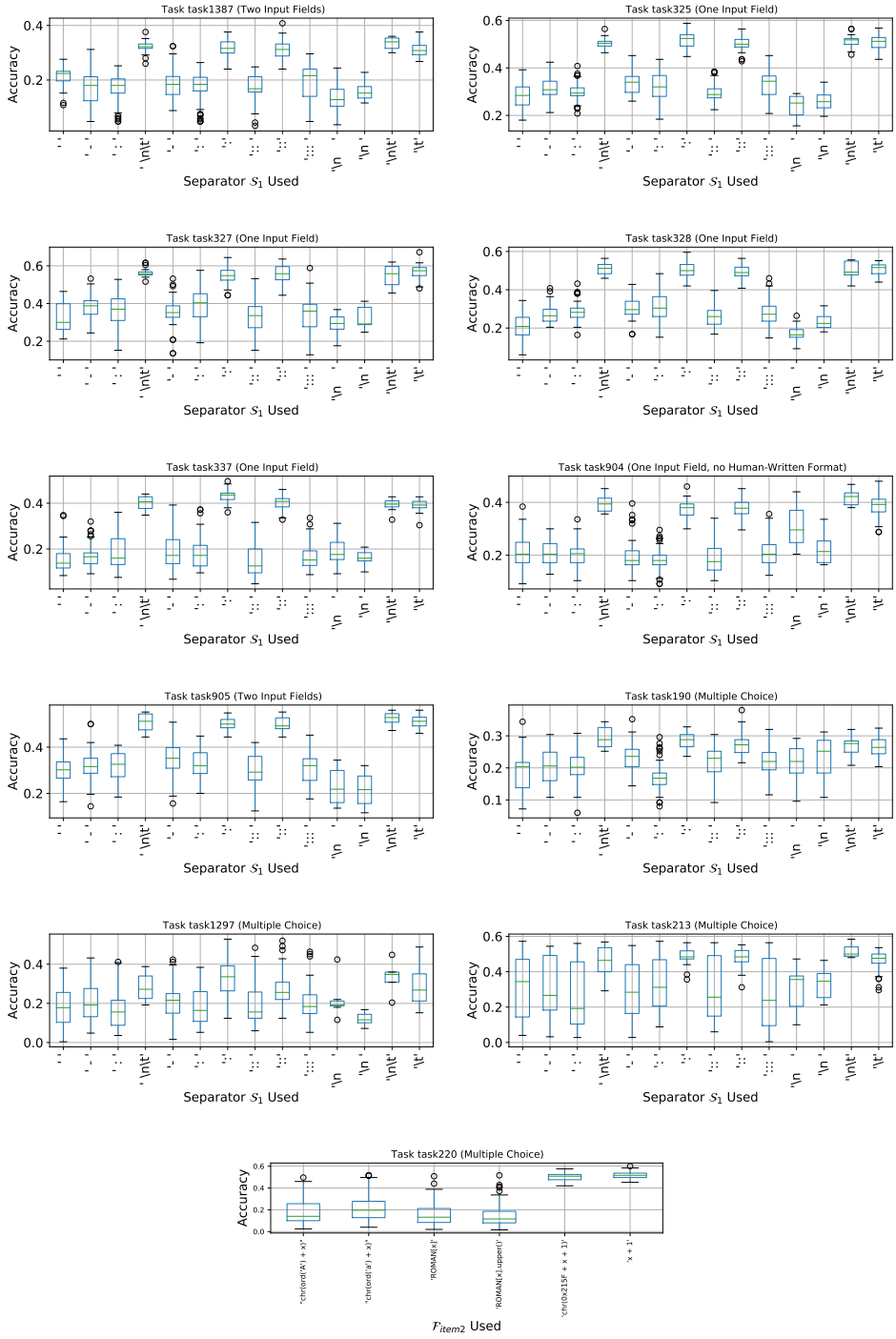

Figure 22: Variance by feature value for Llama-7B 1-shot. Evaluated 31 tasks with 500 formats each, and only plots with a significant difference between values are shown. Exact prefix matching accuracy metric.

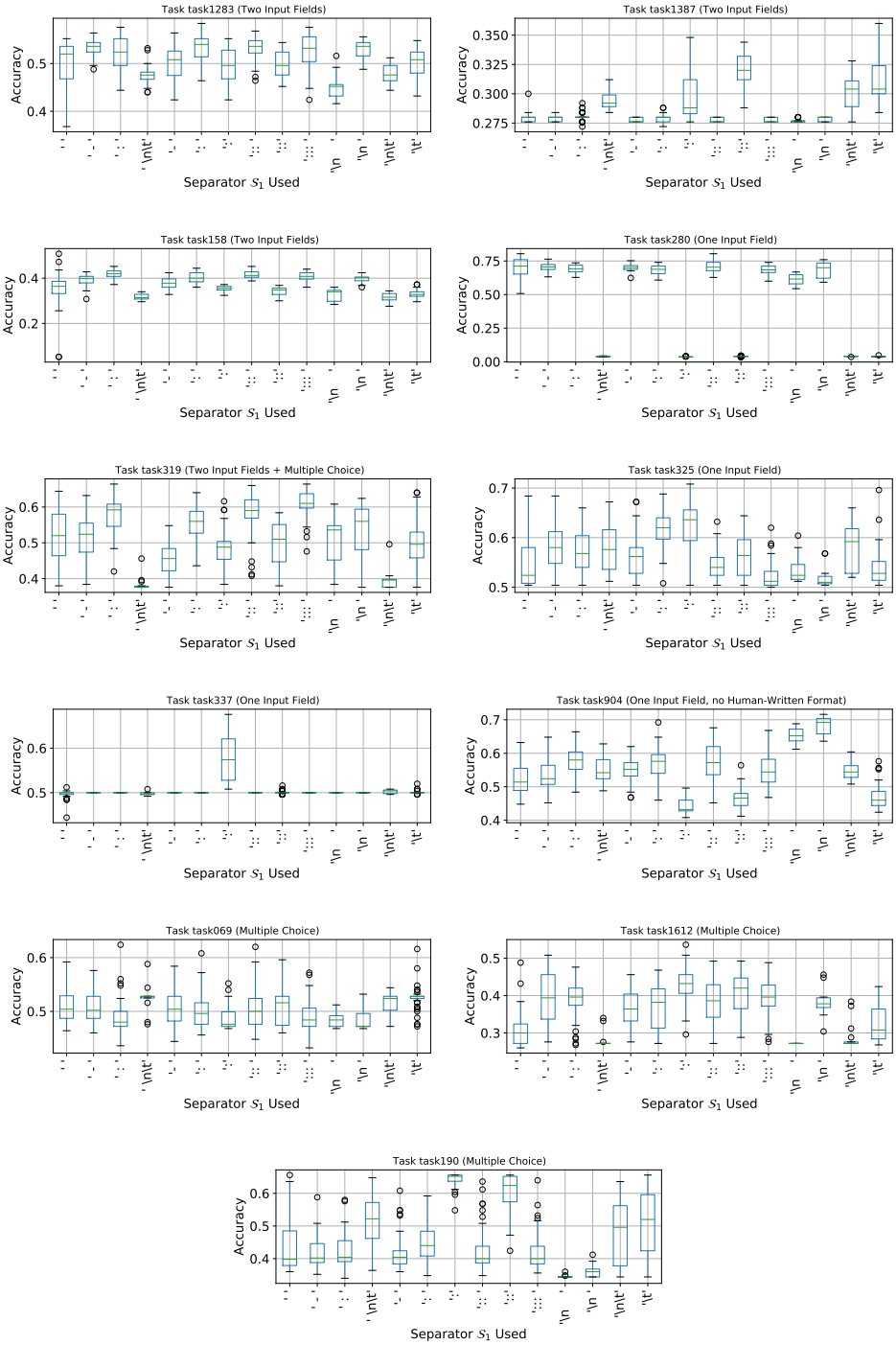

Figure 23: Variance by feature value for Llama-7B 1-shot. Evaluated 31 tasks with 500 formats each, and only plots with strongly significant differences are shown. Option ranking accuracy evaluation metric. Only $\mathcal{S}_1$ boxplots are shown here, see Figure 24 for all others.

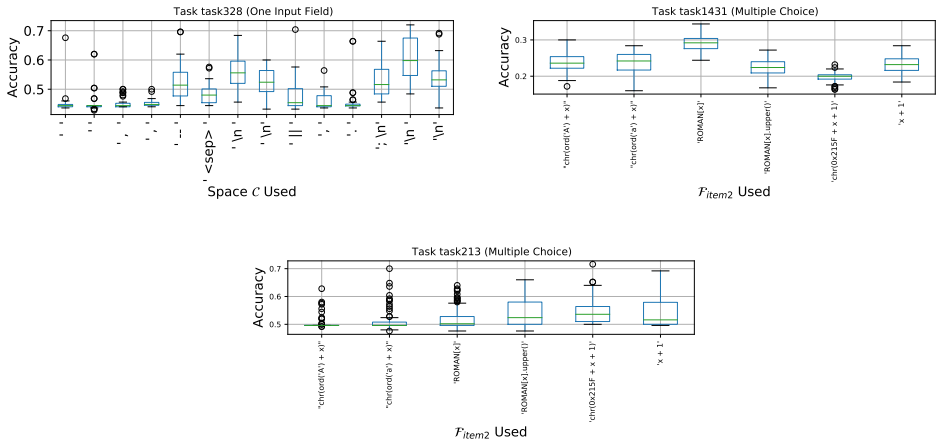

Figure 24: Variance by feature value for Llama-7B 1-shot. Evaluated 31 tasks with 500 formats each, and only plots with strongly significant differences are shown. Option ranking accuracy evaluation metric. See Figure 23 for more plots.

B.5 Thompson Sampling Results

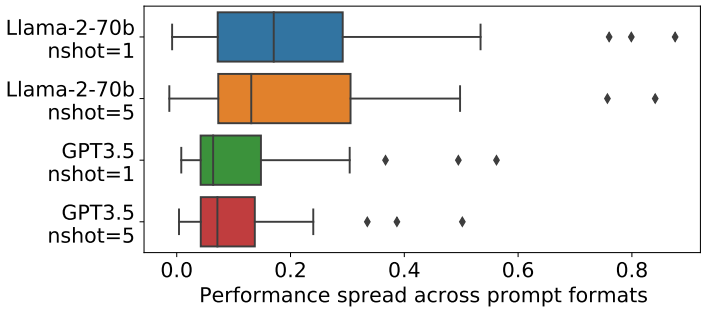

Figure 25: Spreads found using Thompson Sampling for 320 formats, budget of 40000 evaluations. LLaMA-2-70b was evaluated with option ranking, and GPT3.5 with prefix matching given that we cannot access all logits.

Task	Model	Best Format	Worst Format	Best Acc	Worst Acc
task050	Llama-70B	`sentence {} || question {} || answer {}`	`Sentence\t{}\n Question\t{}\n Answer\t{}`	0.62	0.58
task065*	Llama-70B	`Sentence<1>:: {} Sentence<3>:: {} Sentence<4>:: {} Sentence<5>:: {} \nOption A.: {} Option B.: {} \nAnswer\n {}`	`Sentence I)\t{} \n Sentence III)\t{} \n Sentence IV)\t{} \n Sentence V)\t{} Option\t(I)::{}, Option\t(II)::{} Answer {}`	0.88	0.50
task069	Llama-70B	`BEGINNING::: {} MIDDLE A): {} , MIDDLE B): {} ENDING::: {} ANSWER::: {}`	`BEGINNING\n {}\n MIDDLE <I>:: {} MIDDLE <II>:: {}\n ENDING\n {}\n ANSWER\n {}`	0.85	0.58
task070	Llama-70B	`Beginning::: {}; \nMiddle\t1)\t{}; \nMiddle\t2)\t{}; \nEnding::: {}; \nAnswer::: {}`	`Beginning {} || Middle 1.: {} || Middle 2.: {} || Ending {} || Answer {}`	0.80	0.27
task114	Llama-70B	`sentence: {} <sep>answer: {}`	`Sentence: {}\nAnswer: {}`	0.54	0.51
task1186	Llama-70B	`SYSTEM REFERENCE : {}. ORIGINAL REFERENCE : {}. ANSWER : {}`	`System Reference: {}\nOriginal Reference: {}\nAnswer: {}`	0.56	0.51